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Insights into all-vanadium redox flow battery: A case study on components and operational conditions

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ABSTRACT

This works reports the effect of operation conditions on the performance of a 25 cm^2 vanadium redox flow cell (VRFC) featuring commercial carbon felt (CF) as electrode and different commercial membranes, namely electrolyte flow rate and charge-discharge current density. The effect of electrode compression and thermal treatment was also assessed.

The battery performance was evaluated based on the charge-discharge curves and average efficiencies during 20 charge-discharge cycles; the capacity fading was evaluated after 20 cycles. The limiting current was obtained through polarization curves and the ohmic resistance of the battery was obtained by electrochemical impedance spectroscopy (EIS).

The electrodes yielding the highest energy efficiency (EE), *ca.* 85%, at 40 mA cm^{-2} current density were thermally treated at 390 °C or 415 °C during 12 h under air. This result represents an improvement of more than 50% compared with the as-received electrodes. Additionally, energy efficiency increased *ca.* 10% when electrode compression rose from 8% to 25%. Increasing the flow rate from 37.5 cm^3 min^{-1} to 150 cm^3 min^{-1} EE was also enhanced from 76% to 84% at 40 mA cm^{-2}. The type of membrane affects the performance: a comparison between a commercial anion exchange membrane (AEM) and Nafion® series cation exchange membranes (CEM) showed that AEM, generally, yields higher EE than all Nafion® membranes tested; however, the capacity fading of AEM is much higher than CEM.

The information provided in this work shows that, by tuning operational conditions and using suitable components, it is possible to increase the performance of a VRFC without additional costs.

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1. Introduction

In the last decades, a significant increase of energy production based on renewable sources, has been observed, especially solar and wind power. However, photovoltaic electricity (PV) is not dispatchable and its integration in the low voltage grid is not possible above a certain threshold due to its variable generation nature. This threshold dependents of the local grid infrastructure; it was estimated an upper penetration limit of PV generation of about 33% for UK and in the case of Japan this limit is between 5% and 20% [1]. Energy storage systems (ESSs) are then mandatory for managing renewable power supplies, resulting in a more resilient energy infrastructure and cost savings to utilities and consumers.

Among all ESSs, redox flow batteries (RFBs) is one of the most promising technologies for storing large amounts of energy [2,3]. RFB uses redox-solubilized species, which can be stored in external tanks and pumped into the stack as needed. The most appealing aspects of RFB are the scalability, flexibility, high round-trip efficiency, high depth of discharge, long life, fast response and reduced environmental impact [4]. Additionally, the number and size of the cells and the volume of the electrolyte can respectively define power and energy independently. The potential applications of RFBs include load levelling and peak shaving, uninterruptible power supply, emergency backup for hospitals and air-traffic control and integration in wind and photovoltaic energy plants.

Among all RFBs (iron/chromium, vanadium/bromine, bromine/polysulfide, zinc–cerium, zinc/bromine, and all-vanadium), all-vanadium redox flow battery (VRFB) is the most studied and promising chemistry. VRFB exploits the ability of vanadium to exist in different oxidation states reducing, therefore, cross-contamination problems between electrodes. The electrochemical half-cell reactions for charging are as follows:

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$$V^{3+} + e^- \rightarrow V^{2+}\ E^0 = -0.255\ \text{V vs SHE} \quad (1)$$

$$VO^{2+} + H_2O \rightarrow VO_2^+ + 2H^+ + e^-\ E^0 = 1.004\ \text{V vs SHE} \quad (2)$$

The reverse reactions occur during discharge.

Albeit the accomplished progress over the recent years, resulting in several VRFB prototypes worldwide in the megawatt scale [2,5–7], the adoption for lower power range is still significantly hindered by the high capital cost resulting from the modest redox-active materials and low-performance power deliverance. High power densities (operating current density) to reduce the stack size and increasing electrolyte usage (charge-discharge depth) are features to be tackled envisioning the reduction of the capital cost [8].

The most commonly used electrolyte media in VRFBs systems is the pure sulphate, typically 2–5 M H_2SO_4 aqueous solutions. As electrolyte separator, cation exchange membranes (CEMs) such Nafion® are the benchmark membranes for VRFB because of high proton conductivity and good chemical stability [9,10]. Nafion® membrane is, however, expensive and has a significant permeability to vanadium ions [11,12]; several successful approaches have been taken to decrease the permeability to vanadium ions such as the incorporation of nanoparticles [13,14], construction of composites [15], multilayer structuring [16] and surface modification [17,18]. Alternative membranes, less expensive, include nanofiltration membranes [19,20] and anion exchange membranes (AEMs) [21,22]. The latter has received considerable attention for VRFB application mostly due to the low permeability to vanadium ions [22,23].

Porous carbon-based materials (carbon and graphite felts, carbon cloth and carbon paper) are usually used as electrodes due to their stable, three-dimensional network structure, good (electro) chemical stability, high conductivity and relatively low cost. However, electrochemical activity and kinetic reversibility are drawbacks that have to be tackled for improving the battery performance; several approaches have been followed for increasing the electrochemical activity, namely acid treatment [24,25], thermal treatment [26–28], plasma treatment [29], or introducing metal/metal oxide catalysts onto the electrode surfaces [30–34].

Few studies report the effect of VRFB operational conditions, namely electrode compression, flow rate, or current density, on the performance of a VRFB. The typical carbon felt used in VRFBs is rather thick when compared to the electrode of polymer electrolyte fuel cells. The increase of electrode compression (therefore of density of carbon fibres) leads to increased electron transfer but also to a decrease in porosity detrimental to electrolyte transport [35]. Compression, thickness and porosity must be optimized for obtaining the highest current density and the lowest pressure drop.

S.–K. Park et al. [36] reported that the optimum values for capacity (706.5 mA h at 20 mA cm^{-2}) and power (2.77 W at 70 mA cm^{-2}) for an 30 cm^2 active area cell were attained with 30% compression using a 4 mm thick carbon felt; the highest energy efficiency, regardless the current density, was attained with 20% compression. Q. H. Liu et al. [37] studied, among other parameters, the carbon electrode thickness increasing from one to three the number of carbon paper layers obtaining *ca.* 23% increase in peak power density; further increase in the number of layers to 5, 7 and 9 resulted in *ca.* 3%, 13% and 14% decrease in peak power density, respectively. However, the authors did not characterize the charge/discharge and capacity loss of the battery.

M. J. Watt-Smith et al. [38] examined the effect of electrolyte flow rate on a 100 cm^2 cell efficiency and determined that the optimal flow rate was 1.5–2 cm^3 min^{-1} for electrolytes exhibiting different concentrations. The same authors reported also that higher flow rates had a detrimental effect on the efficiencies, particularly in the Coulombic efficiency. Using a 875 cm^2 active area kilowatt class VRFB system, X. Ma et al. [39] reported an optimized strategy for electrolyte flow rate for the charge–discharge cycle: the charge and discharge would start with 10 dm^3 min^{-1} and be increased to 12 dm^3 min^{-1} when the potential reaches 23.1 V (charge mode) and 16.8 V (discharge mode). The system efficiency improved as high as 8% when compared to constant flow rate.

In a more recent study, M. R. Mohamed et al. [40] found that the flow rate has a minor effect on the discharge cell voltage in the range of 60–300 cm^3 min^{-1} when operating with a 25 cm^2 active area cell at 60 mA cm^{-2} of current density. However, for flow rates higher than 300 cm^3 min^{-1}, discharge cell voltage slightly dropped as the reaction is no longer controlled by mass transport. Previously, A. A. Shah et al. [41] arrived to similar conclusions: for a 100 cm^2 active area cell operated at 100 mA cm^{-2}, higher efficiencies could be attained when flow rate increases from 60 to 120 cm^3 min^{-1}; for higher flow rates, the efficiency of the cell is impaired.

Previous studies on the effect of VRFB operational conditions are rarely reported in the literature. This work reports a systematic assessment of operating conditions (charge/discharge current density and electrolyte flow rate) and battery components (electrode compression and membrane) influence on the performance of a lab scale vanadium redox flow cell (VRFC). Additionally, the thermal treatment of the electrodes was assessed and optimized varying treatment time, temperature and atmosphere. The role of the operating conditions and of the thermal treatment of the electrodes on the energy efficiency was assessed based on a detailed characterization of the VRFC, including polarization curves and electrochemical impedance spectroscopy (EIS) analysis. Based on commercial materials, the performance of the VRFB can be promptly and significantly improved with a simple, scalable and inexpensive thermal treatment process of carbon felt electrodes and the suitable selections of operating conditions and cell components.

2. Experimental

2.1. Materials

Anion exchange membrane (AEM) FAP 450 was purchased from Fumatech (BWT AG, Germany) whilst cation exchange membranes (CEM) Nafion® 212, 115 and 117 membranes, denoted as N212, N115 and N117 respectively, were purchased from Quintech (Germany). CEMs were kept in 1 M H_2SO_4 solution for 24 h and then stored in deionized water before further use. The as-received AEMs were used without any treatment according to the supplier's recommendation. Between experiments, all membranes were kept in 1 M H_2SO_4 solution at 80 °C until all vanadium stains were removed and stored in deionized water before being used again.

PAN-based carbon felt (CF) with 6 mm of uncompressed thickness was used as electrode material (95% uncompressed porosity, Chemshine Carbon, China).

The electrolyte consisted of a commercially available 4 M sulfuric acid-based vanadium solution (1.6 M V^{3+}/V^{4+} 50:50 M ratio, GfE Metalle und Materialien GmbH, Germany) and it was used without further treatment or purification.

2.2. Carbon felt electrodes pre-treatment

All electrode samples were boiled in 1 M H_2SO_4 solution to remove fat and non-carbon coating and thoroughly rinsed with deionized water. Afterwards, the electrode samples were dried overnight at 80 °C and stored until further use. The CF electrodes

samples were thermally treated under air atmosphere at in the range of 340–440 °C during 2 h–24 h using a furnace muffle. For assessing the effect of the atmosphere during thermal treatment on carbon felt fibres, an inert atmosphere composed of 100% N_2 was also used. To assess the amount of oxide functional groups attached to the carbon felt electrodes surface thermogravimetric analysis, TGA, was carried out in a Netzsch TG 209 F1 Iris instrument. The samples, under nitrogen atmosphere, were heated from room temperature to 900 °C at 3 °C min^{-1}.

2.3. VRFC setup and performance

Fig. 1 illustrates half of the single cell VRFC used in this study. The cell includes two stainless steel endplates (each $120 \times 100 \times 15\ mm^3$), two copper plates (each $90 \times 70 \times 1\ mm^3$) and two graphite foils ($90 \times 70 \times 0.5\ mm^3$) as current collectors. The electrical insulation from the endplates was ensured with 2 mm thick LDPE (low-density polyethylene) separators.

Each electrode has a projected geometric area of $50 \times 50\ mm^2$ and were compressed up to 25% of its original thickness, unless otherwise stated. To ensure the desired compression and for preventing leakage of electrolyte EPDM (ethylene propylene diene monomer) rubber gaskets were properly placed between parts. The electrolyte was recirculated from the tanks to the cell using a two-headed (314D) Watson-Marlow 323 peristaltic pump featuring Verdeflex® tubing (4.8 mm inner diameter). This type of tubing minimizes the permeation of oxygen from atmospheric air and the effect of the acidic media on the tubing service life. The flow rate ranged between 37.5 and 300 $cm^3\ min^{-1}$ (corresponding to linear flow velocity past the electrode of 0.32–2.52 cm s^{-1}) depending on the specific conditions of the test as described in each discussion sub-section. 70 cm^3 of electrolyte were used in each tank and the system was purged with nitrogen throughout the experiments to prevent oxidation of vanadium redox species by air.

The electrochemical measurements, charge and discharge cycle, polarization curves and electrochemical impedance spectroscopy (EIS), were carried out at room temperature using an electrochemical workstation (EW) Zahner Zennium coupled to a PP240 (Zahner Elektrik).

2.4. Charge discharge cycles

Unless otherwise stated, the cell was cycled at 40 mA cm^{-2} of current density and cut-off values for the open-circuit cell voltage of 1.5 V during charge, OCV_c, and 1.2 V during discharge, OCV_d were used to define the charge and discharge times, respectively. These specifications ensured that the cell was operated between 10% and 90% of state of charge (SOC) preventing the evolution of O_2 and H_2. For each set of operational conditions, the cell underwent 20 cycles of charge/discharge to assess the capacity fading. The coulombic efficiency (CE), voltage efficiency (VE), energy efficiency (EE) and discharge capacity (C_d) of the cell were calculated from equations [42]:

$$\text{Coulombic efficiency (CE)}: \frac{\int_0^t I_d dt}{\int_0^t I_c dt} \tag{3}$$

$$\text{Voltage efficiency (VE)}: \frac{\int_0^t E_{cell,\ d} dt}{\int_0^t E_{cell,\ c} dt} \tag{4}$$

$$\text{Energy efficiency (EE)} = \text{CE} \times \text{VE} \tag{5}$$

$$\text{Discharge capacity in mA h (Cd)} = \frac{\int_0^t I_d dt}{3600} \tag{6}$$

where I denotes current, E_{cell} is the cell potential difference and subscripts c and d denote charging and discharging phases, respectively.

To ensure identical initial conditions, fresh electrolyte, newly prepared electrodes and clean membranes were used to make accurate comparisons between different tests. Table 1 summarizes all the experimental conditions.

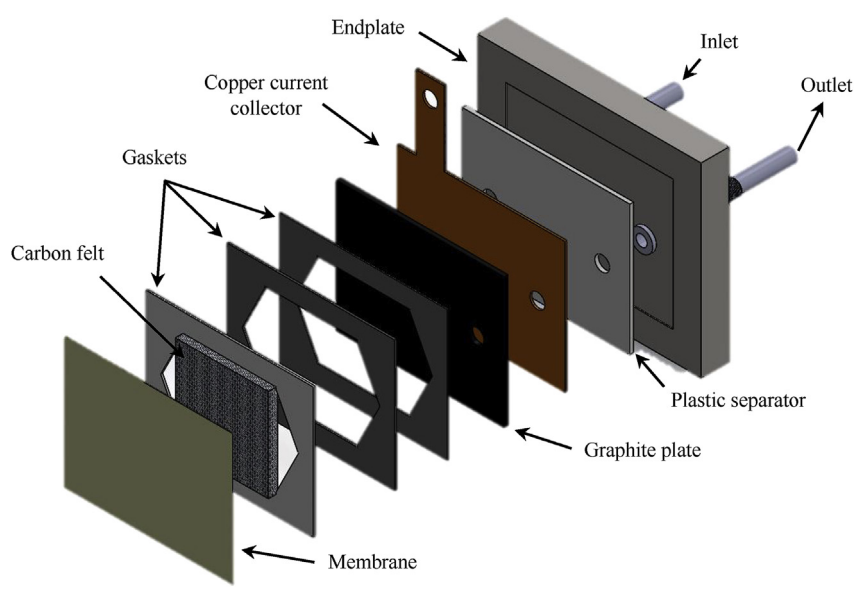

Fig. 1. Scheme of the VRFC single cell used in this work – cathode side.

Table 1
Experimental conditions employed in the study of the VRFC performance.

j/mA cm^{-2}	Q/cm^3 min^{-1}	Compression (%)	V_{range}/V	Thickness/mm	Membrane	Thermal treatment t/h	T/°C
40	150	8 17 25 33	1.60–0.80	6.0	FAP 450	6	390
40	35 75 300	25	1.60–0.80	6.0	FAP 450	6	390
80 120 160	300	25	1.60–0.80 1.70–0.80 1.75–0.80	6.0	FAP 450	6	390
40	75	25	1.60–0.80	6.0	FAP 450	2 12 24	390
40	75	25	1.60–0.80	6.0	FAP 450	12	340 415 440
40	150	25	1.60–0.80	6.0	N212 N117 N115 FAP 450	12	415
40	225	25	1.65–0.80	4.6 6.0	FAP 450	12 h	415

2.5. Polarization curves and electrochemical impedance spectroscopy

The discharge polarization curves were obtained from fully charged state down to 0.8 V at 10 mV s^{-1} of discharge rate employing potentiostat control. A steady current below 20 mA at a cell potential of 1.60 V was used as indication of full charge.

Electrochemical impedance spectroscopy (EIS) was used to obtain the ohmic resistance of the cell, R_{ohm}, and the charge transfer resistance of the electrodes, R_{ct}. R_{ohm} was obtained when the Nyquist plot crossed the real axis at high frequencies (at *ca.* 20 kHz) while the R_{ct} was obtained from diameter of the semicircle of the Nyquist diagram. EIS was performed at null current in a frequency range between 100 mHz and 50 kHz with an AC amplitude of 5 mV. Before starting the EIS measurements, the cell was charged at the desired state of charge (SOC).

All data related to polarization curves and Nyquist plots are included in Appendix A.

3. Results and discussion

3.1. Carbon felt thermal treatment

To enhance the performance of the carbon felt a simple, low-cost and easily scalable thermal treatment was performed. The treatment modifies the surface properties of the carbon material, increasing the wettability and the electrochemical activity by introducing surface functional groups (SFG) [28].

The effect of the thermal treatment, performed during 12 h at temperatures between 340 and 440 °C, and at constant 390 °C during 2–24 h, was evaluated performing 20 charge-discharge cycles at 40 mA cm^{-2} and 75 cm^3 min^{-1} (mean linear flow velocity of 0.64 cm s^{-1}). According to the literature, around 400 °C is the temperature at which the maximum amount of oxygen is chemisorbed on the surface of the carbon material [28,43,44]. Fig. 2 shows the history of the second charge/discharge cycle and the average efficiencies for the 20 cycles and discharge capacity loss (DCL), between cycle #20 and cycle #2, for different thermal treatment durations at 390 °C (Fig. 2a and b) and temperatures (Fig. 2c and d) during 12 h treatment. The second cycle is representative of the 20 charge/discharge cycles without the influence of capacity fading.

Coulombic and voltage efficiencies are, in general, the properties that better evaluate the performance of a VRFB. From coulombic efficiency (CE) is possible to assess the electron transport into and out of the cell and to track side reactions and/or other coulombic losses. From voltage efficiency (VE) is possible to assess losses through overpotential and electrolyte crossover.

The time of charge/discharge cycle dramatically increased after employing the thermal treatment at 390 °C, Fig. 2a, resulting in an increase of 2.7–3.4-fold in the discharge capacity of the battery, depending on the duration of the thermal treatment (2 h–24 h), when compared to the capacity provided by the untreated electrodes (685 mA h). The capacity values obtained for 12 h and 24 h of treatment are very close, 2307 and 2327 mA h, respectively, showing that the longer treatment is not translated in a noticeable cell performance. Additionally, treating the electrodes during 24 h led to a higher value of capacity fading of the battery when compared to a 12 h treatment, Fig. 2b. Therefore, 12 h was chosen as the optimum thermal treatment duration when varying the temperature (Fig. 2c); the storage capacity increased for the electrode treated at 415 °C when compared with 390 °C, as a result of increasing of the charge/discharge cycle time: 2307 mA h and 2321 mA h, respectively. Electrodes treated at 340 °C and 440 °C, yielded 1917 mA h and 1840 mA h, respectively.

The time or temperature of carbon felt thermal treatment has little influence on the CE but affects VE, Fig. 2b and d. Increasing the time of thermal treatment from 2 h to 12 h VE increased linearly from 76% to 86%. Increasing thermal treatment duration to 24 h led to a VE increase of a mere 2% when compared to 12 h, Fig. 2b. Increasing the temperature of the thermal treatment, from 340 °C to 390 °C or 415 °C, the VE increased from 79% to 86% and 87% respectively, Fig. 2d. The highest EE, 85%, was obtained when the battery was featuring carbon electrodes treated for 12 h at 390 °C or 415 °C and for 24 h at 390 °C, being in accordance to the literature in terms of temperature of treatment [28,29,37]. The improvement of the electrochemical activity of the electrodes after thermal treatment has been ascribed to the amount and nature of SFG and to morphological defects on the carbon fibres providing more redox reaction sites [27–29,37]. On the other hand, when temperature increased from 415 °C to 440 °C, EE drastically decreased from 85% to 78%, most likely due to the degradation of the electrode fibres at

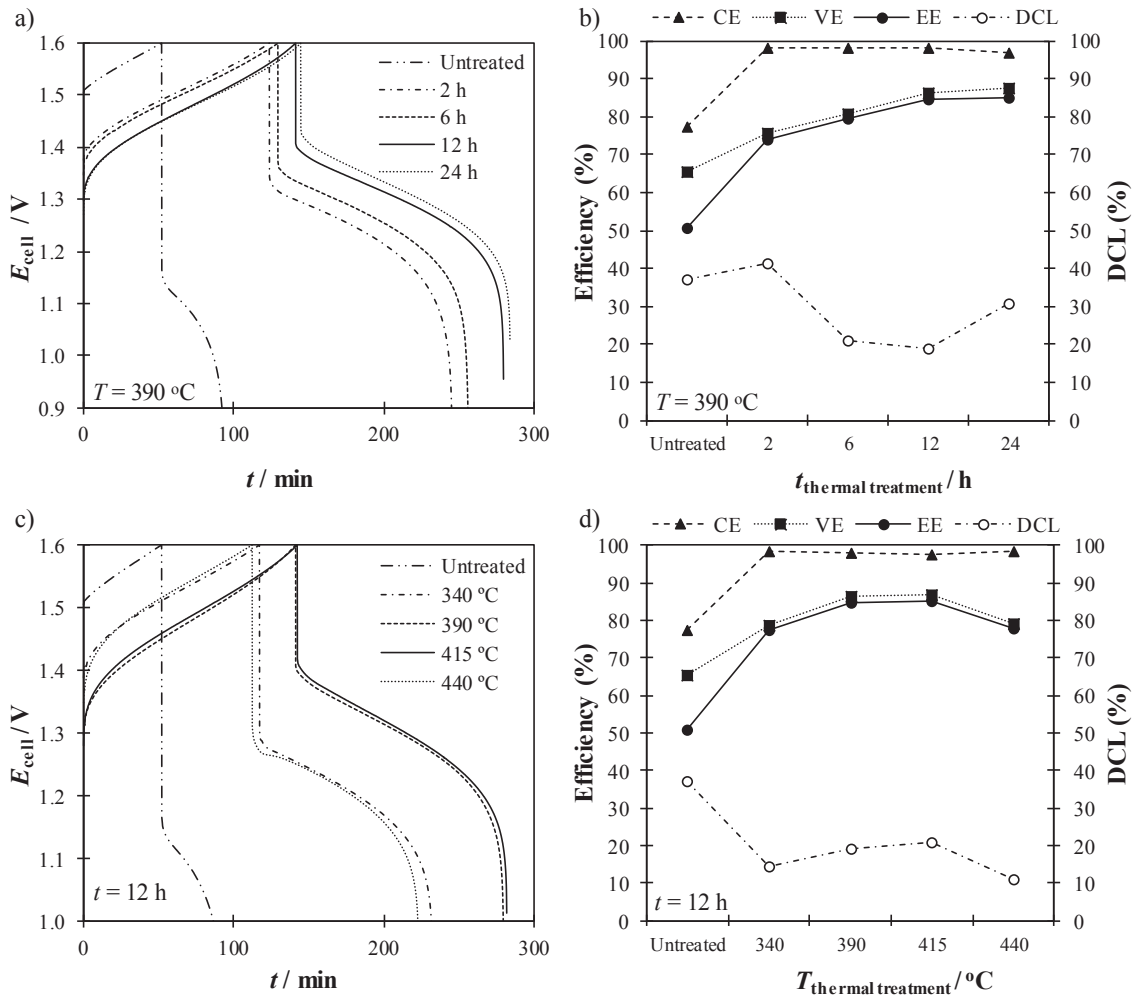

Fig. 2. Effect of thermal treatment duration, $t_{\text{thermal treatment}}$, on: (a) the history of the second charge/discharge cycle; (b) average CE, VE, EE and DCL for 20 cycles; Effect of thermal treatment temperature, $T_{\text{thermal treatment}}$, on: (c) the history of the second charge/discharge cycle; (d) average CE, VE, EE and DCL for 20 cycles. Electrode compression = 25%; Membrane: FAP 450; $Q_{\text{electrolyte}} = 75\ \text{cm}^3\ \text{min}^{-1}$; $j = 40\ \text{mA cm}^{-2}$; $\text{OCV}_d = 1.2$ V, $\text{OCV}_c = 1.5$ V.

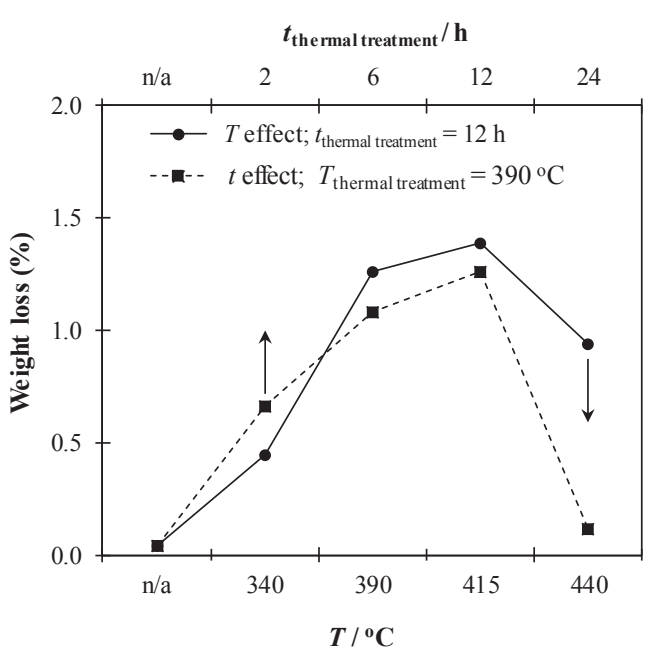

Fig. 3. Effect of treatment time and temperature on weight loss of carbon felt samples under nitrogen atmosphere.

higher temperatures. It is important to note that increasing the time of treatment the amount of SFG will also increase.

Regarding the DCL (discharge capacity loss), it is clear that 6 and 12 h of treatment at 390 °C provided the lowest values, *ca.* 1% cycle^{-1} (Fig. 2b). On the other hand, fixing 12 h of thermal treatment, 340 °C and 440 °C yielded the lowest values of DCL, 0.7 and 0.5% cycle^{-1} (Fig. 2d). Like Li-ion batteries, where intercalation of Li inside the structure of the electrodes has an important role to prevent capacity loss [45], the intercalation of SO_4^{2-} ions in the carbon felt electrodes can be an important factor. This hypothesis was proposed by A. Moissette et al. [46] and later by I. Derr et al. [47]. The intercalation of SO_4^{2-} ions combined with applied potential and the recirculation of electrolyte would cause a considerable mechanical stress in the electrode leading to its degradation. Taking this into account, 12 h of treatment at 390 °C or 415 °C may enable a more effective intercalation of SO_4^{2-} ions combined with an optimal amount of surface oxide species leading to an enhancement of VRFC performance. By imposing 12 h of thermal treatment at 340 °C and 440 °C a better intercalation of SO_4^{2-} ions in the electrode structure seems to happen as DCL decreases in both cases. Regarding the untreated electrode, intercalation of SO_4^{2-} ions (considering the hydrophobic properties of this material) and deposition of oxide groups may be hard to occur yielding higher values of DCL. It should be pointed out that the DCL value found for 2 h of thermal

treatment at 390 °C is higher in comparison to the untreated sample (Fig. 2b). In this case, 2 h may not be enough time to make an effective addition of SFG nor the necessary morphological changes to enable the intercalation of SO_4^{2-} ions, despite an increase in the efficiencies.

To access the presence of SFG, the electrode weight loss after thermal treatment was obtained using thermogravimetric analysis, Fig. 3. The measurements were performed under nitrogen atmosphere up to 900 °C at 3 °C min^{-1}; the weight loss was attributed to the release of the SFG in the form of carbon dioxide, CO_2, or carbon monoxide, CO. The highest weight loss was observed for the electrodes treated during 12 h at 390 °C and 415 °C, namely 1.3% and 1.4%, respectively. Thermally treating carbon felt electrodes during 24 h at 390 °C or at 440 °C did not result in an increase of SFG when compared to samples treated during 12 h at 390 °C or at 415 °C for 12 h. It is possible that a prolonged time of exposure as well as an exposure to high temperature may induce the decomposition of the electrode surface with evolution of CO_2 or CO [28].

EIS and polarization curves were obtained to further investigate the effect of time and temperature of thermal treatment on ohmic resistances, R_{ohm}, charge transfer resistances, R_{ct}, and limiting current of the battery - Fig. 4. R_{ohm} include resistance ionic transport through the electrolyte (electrolyte solution and membrane), electrical resistance in the electrodes and contact resistance between cell components while R_{ct} results from charge transfer reactions at the interface between the electrode and electrolyte.

R_{ct} decreases drastically after employing thermal treatment (Fig. 4a and c). The gradual decrease as the duration of thermal treatment increases indicates that electrochemical activity of the electrodes increased – Fig. 4a. The lowest R_{ct} value was found for 24 h thermal treatment – 0.29 Ω cm^2. On the other hand, increasing the temperature R_{ct} follows an opposite trend, *i.e.*, it gradually increases and the highest R_{ct} value was obtained for 440 °C (2.42 Ω cm^2). Regarding R_{ohm}, 2 h at 390 °C (Figs. 4a) and 12 h at 340 °C yielded the lowest values, 2.11 Ω cm^2 and 2.37 Ω cm^2, respectively. The highest R_{ohm} values were found for 12 h at 390 °C (Figs. 4a) and 12 h at 415 °C (Fig. 4b). From Fig. 4b and d, it is also possible to observe that employing 12 h at 390 °C or 12 h at 415 °C, the limiting currents were the lowest: 80 mA cm^{-2} and 45 mA cm^{-2} respectively. The highest limiting current density values were found for 2 h at 390 °C and for 12 h at 340 °C. Results suggest that time may increase the amount of SFG inserted in the bulk of the electrode, while temperature should increase the amount of surface defects on the fibres increasing the surface area available for fixing SFG [29,48]. Notwithstanding, as the amount of oxygen species effectively increases in the porous media so does the ohmic

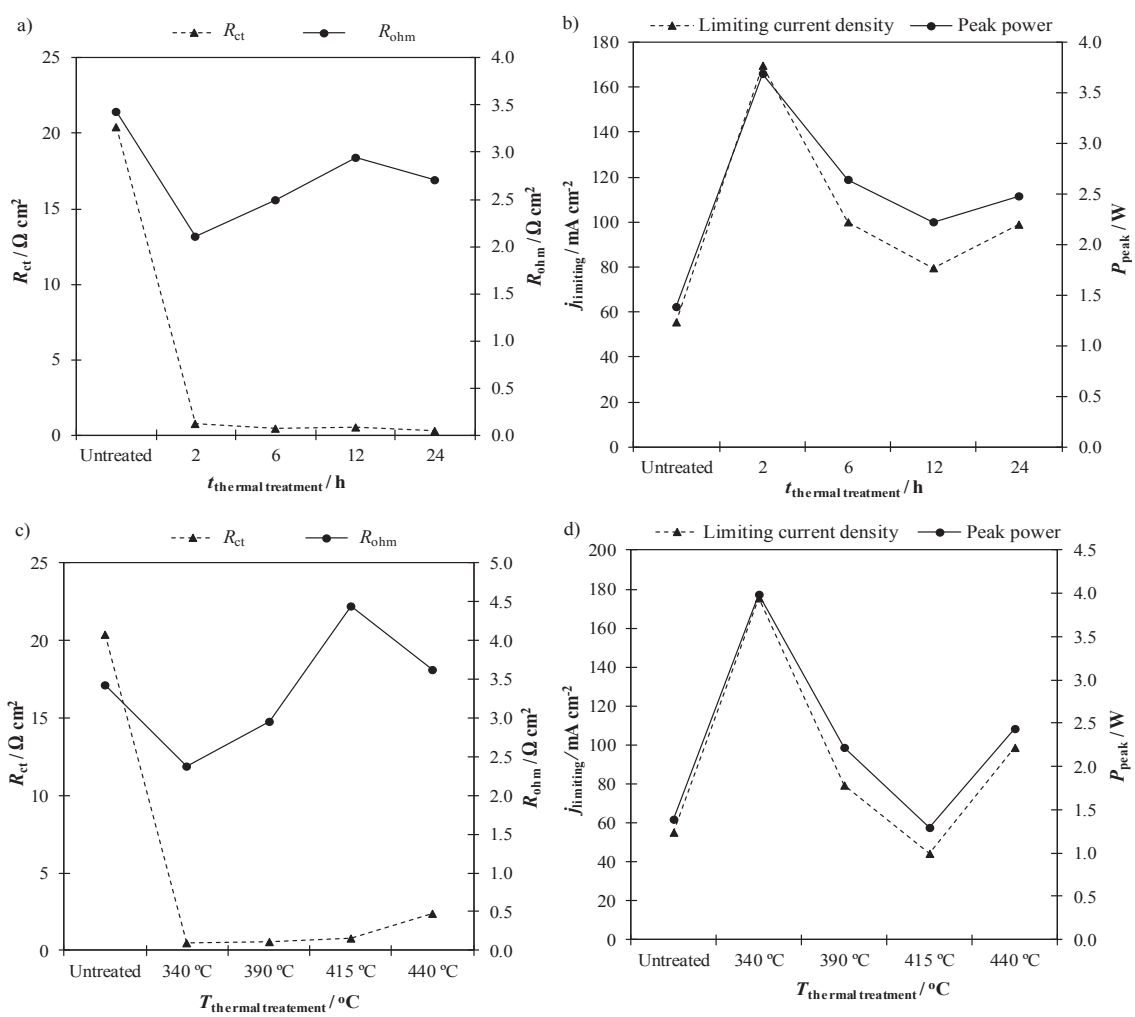

Fig. 4. Effect of thermal treatment duration, $t_{thermal\ treatment}$, on: (a) R_{ct} and R_{ohm} at 50% SOC; (b) Limiting current density, $j_{limiting}$, and peak power, P_{peak}, between 1.6 V and 0.8 V; Electrodes treated at 390 °C. Effect of thermal treatment temperature, $T_{thermal\ treatment}$, on: (c) R_{ct} and R_{ohm} at 50% SOC; (d) Limiting current density, $j_{limiting}$, and peak power, P_{peak}, between 1.6 and 0.8 V; Electrodes treated for 12 h. Electrode compression = 25%; Membrane: FAP 450; $Q_{electrolyte}$ = 75 cm^3 min^{-1}.

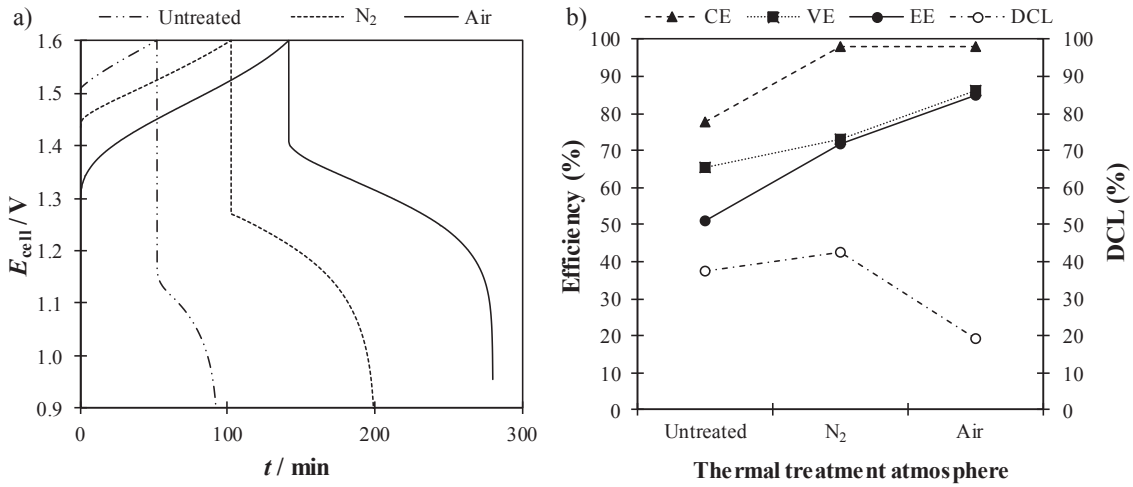

Fig. 5. Effect of thermal treatment atmosphere on: (a) history of the second charge/discharge cycle; (b) CE, VE, EE and DCL after 20 cycles. Electrodes treated at 390 °C for 12 h; Electrode compression = 25%; Membrane: FAP 450; $Q_{electrolyte} = 75\ cm^3\ min^{-1}$; $j = 40\ mA\ cm^{-2}$; $OCV_d = 1.2\ V$, $OCV_c = 1.5\ V$. Electrodes treated at 390 °C for 12 h.

and mass transport resistances. 12 h or 24 h at 390 °C or 12 h at 415 °C seems to be the optimal duration and temperature values, respectively, for effectively inserting oxygen species in the bulk of electrodes and for providing the highest VRFC performance as evidenced in Fig. 2b and d.

In Fig. 5 is depicted the effect of the atmosphere on the thermal treatment of electrodes. As previously mentioned, the introduction of SFG has a significant role in the charge/discharge cycle time and therefore in the VRFC storage capacity (Fig. 5a): the battery comprising electrodes treated with atmosphere air displayed an increased capacity, *ca.* 42% from 1629 to 2307 mA h, compared to electrodes treated with N_2. In terms of efficiencies and capacity loss - Fig. 5b – treating electrodes in air also yielded higher values of EE: 85% when treated in air against 73% under N_2 atmosphere. The difference between the charge and discharge potentials is lower for electrodes treated with air, suggesting that VRFC performance is highly dependent on SFG formed during the thermal treatment under air. An interesting observation is the decrease of the DCL (discharge capacity loss) from 2.1% $cycle^{-1}$ to 0.95% $cycle^{-1}$ when the treating atmosphere changed from N_2 to air. Probably, the presence of SFG not only catalyse the vanadium redox reactions but also delays the fading of capacity hindering the O_2 and H_2 gases evolution. It is worth noting that carbon felt electrodes may undergo chemical aging by mere contact with electrolyte [47], which would justify the capacity loss even for the untreated electrodes.

SEM analysis carried out on samples shows the surface

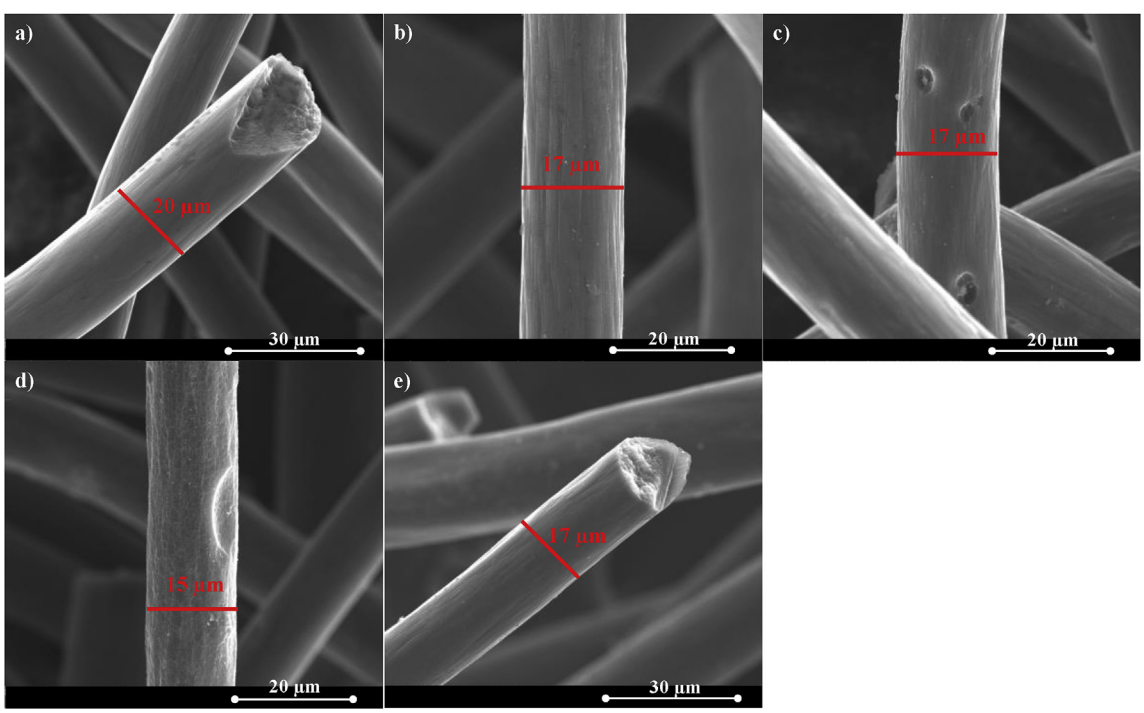

Fig. 6. SEM micrographs of carbon felt samples: a) untreated; b) treated in air at 340 °C for 6 h; c) treated in air at 390 °C for 6 h; d) treated in air at 440 °C for 6 h; e) treated in N_2 at 390 °C for 6 h.

Table 2
Properties of the carbon felt electrodes used in this work.

Compression (%)	Thickness/mm	Porosity[a] (%)	Q/cm^3 min^{-1}	v^b/cm s^{-1}
8	5.5	90.7	150	1.00
17	5.0	89.1		1.12
25	4.5	88.4		1.26
33	4.0	86.1		1.44

[a] Porosity was calculated by interpolation from S.–K. Park et al. [36].
[b] v is the mean linear flow velocity past the electrode surface.

morphology before (Fig. 6a) and after thermal treatment (Fig. 6b–d) under air atmosphere. The influence of nitrogen atmosphere during the thermal treatment is also shown (Fig. 6e). Thermally treating the samples induced a reduction in the diameter of the fibre regardless the atmosphere used. However, when air is used it is possible to observe changes on the surface of the fibres as craters start to appear ($T = 390$ °C). At 440 °C the surface of the fibre seems to be thinner and delaminated justifying the fact these samples become very friable. Regarding the treatment under nitrogen, Fig. 6e shows no significant changes compared to Fig. 6b suggesting that the surface morphology changes after treatment is not the only source of improvement, as the effective insertion of oxygen functional groups in the bulk of the electrodes plays a critical role on the energy performance.

For the remaining studies, the carbon felt electrodes were treated at 390 °C for 6 h or at 415 °C for 12 h.

3.2. Influence of the carbon felt compression

In this section, carbon felt electrodes were thermally treated in air at 390 °C for 6 h and the VRFC performance was accessed incorporating these electrodes with different rates of compression (between 8% and 33%). Although the same flow rate of 150 cm^3 min^{-1} was used, different linear velocity was obtained as the thickness and porosity of the electrode changes with compression. The determination of porosity, listed in Table 2, was interpolated from the work developed by S.-K. Park et al. [36] given the similarities of experimental conditions and to the fact that no experimental determination of porosity was possible at the time of this study.

Regarding the VRFC performance upon different compression, Fig. 7a shows the second of 20 cycles performed under 40 mA cm^{-2} of current density. Increasing the compression in the range of 8%–25%, charge and discharge cycle time and storage capacity increased: 8% compression yielded 1587.6 mA h increasing up to 2333.5 mA h for 25% compression. It is known that the catalytic activity of an electrode depends on its surface morphology and charge/mass transport properties are related to its macro-scale porous structure [49]. Porosity is, in fact, a key parameter affecting the electrolyte flow pattern since it determines the effective transport area. Increasing the compression up to 25% the flow distribution might be improved. Using a higher compression (33%) the storage capacity decreased to 2203 mA h, which is lower than the value observed for 25% of compression or even that of 17% compression. Very high compressions may improve the electrical contact between the components (carbon plate/electrode) and consequently the electronic conductivity however, the electrolyte flow pattern and ionic transport through the electrode may be affected.

Fig. 7b depicts the average CE, VE, EE and DCL for different compression ratios for 20 cycles of charge and discharge. The CE increases from 89% up to 98% when compression increases from 8% to 17% being ascribed to a better distribution of the electrolyte through the electrode as well as to a higher electronic conductivity at the interface between the bipolar plate and carbon felt. Increasing further the compression to 25% and to 33% CE stays almost constant (*ca.* 97%). On the other hand, VE is constant between 8% and 25% of compression (*ca.* 86%), slight decreasing to 84.5% at 33% of compression. The decrease in the VE may be ascribed to the fact 33% compression induces linear flow velocities that no longer control the reaction at electrode/electrolyte interface. Less effective distribution of electrolyte in the electrode and lower residence time will increase limitations on mass transport leading to a lower energy efficiency of the cell.

In addition, DCL was assessed for different compressions: 17% of compression provided 1.4% of capacity loss per cycle whilst 8% compression showed 4.2% loss each cycle. The latter is the highest value and the former the lowest among all compressions tested. 25 and 33% compression led to *ca.* 1% cycle^{-1} of DCL. The capacity fading during cycling may be due to the differential rates of the gaseous evolution and the different permeability of the vanadium ion, which can lead to an imbalance of state-of-charge (SOC) between the two half-cell electrolytes [50]. The previous hypothesis regarding intercalation of SO_4^{2-} in electrode structure can also

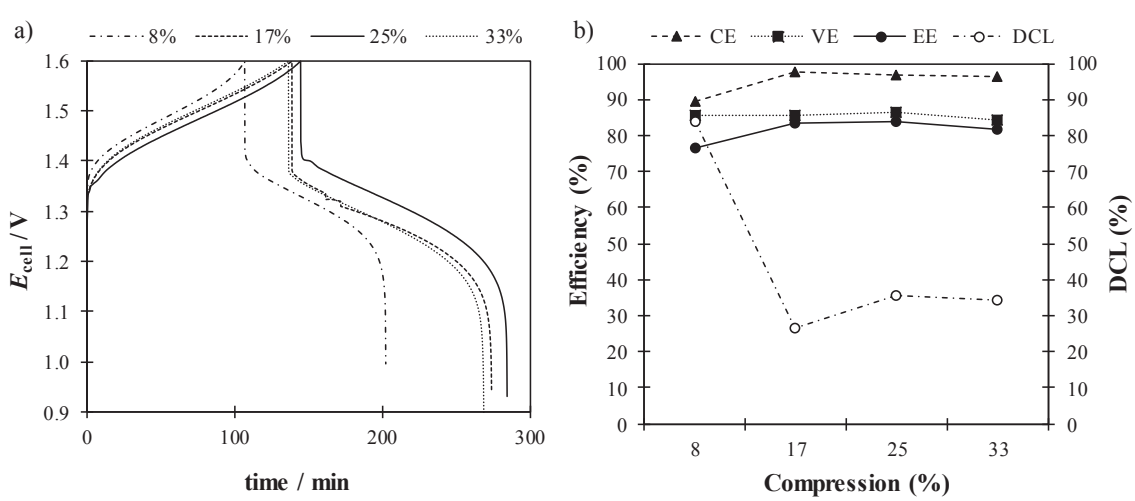

Fig. 7. Effect of electrode compression on: (a) history of the second charge/discharge cycle; (b) average CE, VE, EE and DCL for 20 cycles. $Q_{electrolyte} = 150$ cm^3 min^{-1}; Electrodes treated in air atmosphere at 390 °C for 6 h; Membrane: FAP 450; $j = 40$ mA cm^{-2}; $OCV_d = 1.2$ V, $OCV_c = 1.5$ V.

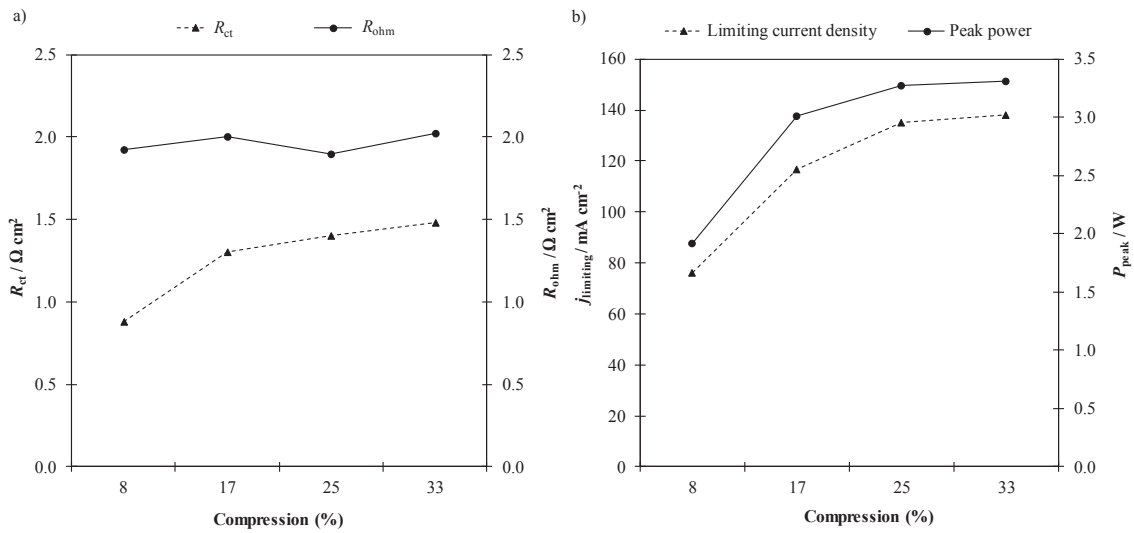

Fig. 8. Effect of electrode compression on: (a) R_{ct} and R_{ohm} at 50% SOC; (b) limiting current density, $j_{limiting}$, and peak power, P_{peak}, between 1.6 and 0.8 V. $Q_{electrolyte}$ = 150 cm^3 min^{-1}; Electrodes treated in air atmosphere at 390 °C for 6 h; Membrane: FAP 450.

justify the large difference between the results obtained for 8% of compression and the other compression values: using such low compression, full de-intercalation from electrode structure of SO_4^{2-} may occur leading, this way, to a higher loss of capacity.

Fig. 8 shows the effect of compression on R_{ohm}, R_{ct} and on the limiting current density. In opposition to the results by S.–K. Park et al. [36] regarding R_{ohm} as a function of compression, Fig. 8a shows that R_{ohm} variation was negligible – in all cases R_{ohm} was *ca.* 2 Ω cm^2 – while R_{ct} increases logarithmically between the compression values of 8% and of 33%. The increasing of R_{ct} with the compression indicates that compression has a greater impact on the kinetics of the reaction than on the ionic or on the contact resistances. This trend reflects on the limiting current density as it also increases logarithmically from 76 mA cm^{-2} to 138 mA cm^{-2} with the compression (corresponding to a peak power between 1.9 W and 3.3 W) – Fig. 8b. The logarithmic shape of the limiting current density was assigned to a higher mass transport of reactants with compression (between 8% and 25% of compression corresponding to a mean linear flow velocity in the range of 1.00–1.26 cm s^{-1}) due to a better electrolyte distribution leading to a non-polarization of active species. However, for the highest compression tested (corresponding to a mean linear flow velocity of 1.44 cm s^{-1}) the limiting current density did not change, which means that the reaction may no longer be controlled by the mass transport. Probably, 33% of compression does not provide a good electrolyte distribution nor the necessary residence time to the reaction take place. The combined effect of charge transfer and of the mass transport resistances increase when the compression increases may justify the slight decrease in the VRFC performance above 25% of compression.

These results are in line to those reported in a study conducted by S.–K. Park et al. [36]. These authors found that a 30 cm^2 cell with a 30% compression electrode (between 0% and 30% of compression) yielded the highest values of capacity (706.5 mA h at 20 mA cm^{-2}) and power (2.77 W at 70 mA cm^{-2}). The highest energy efficiency was, however, attained with 20% compression due to the combined effects of ohmic resistance, electrolyte transport property, and longer charge/discharge time [36].

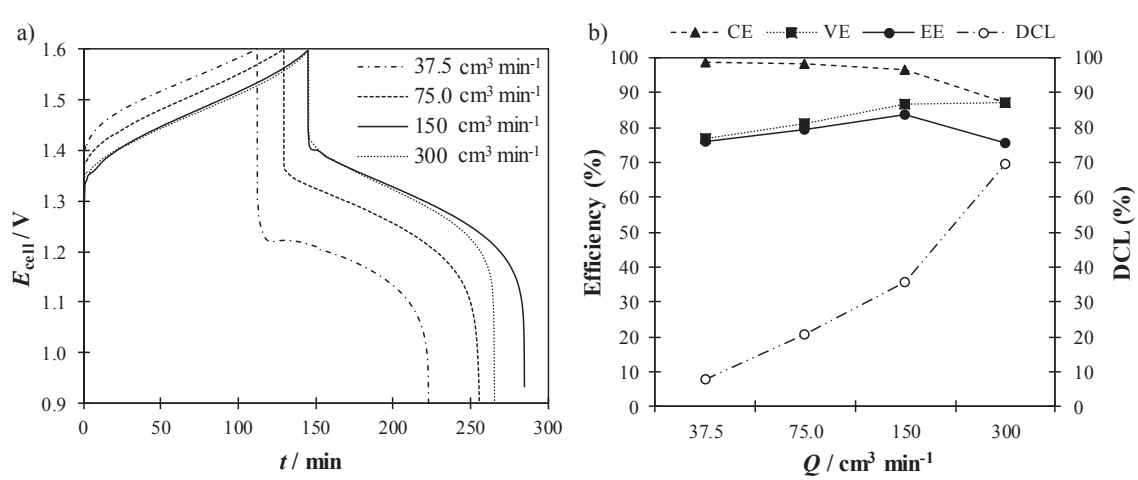

Fig. 9. Effect of $Q_{electrolyte}$ on: (a) history of the second charge/discharge cycle; (b) average CE, VE, EE and DCL for 20 cycles. Electrode compression = 25%; Electrodes treated in air atmosphere at 390 °C for 6 h; Membrane: FAP 450; j = 40 mA cm^{-2}; OCV_d = 1.2 V, OCV_c = 1.5 V.

3.3. Influence of the feed flow rate

An efficient way of reducing the mass transport resistance and achieving higher cell energy efficiencies is employing higher flowrates [51]; however, the overall EE should also balance the increase in mechanical energy for pumping the electrolytes during the charging/discharging cycles.

Fig. 9 illustrates the effect of varying the feed flow rate on the history of the second cycle of charge/discharge (Fig. 9a) and on CE, VE, EE and, DCL (Fig. 9b). For this set of experiments, 25% of compression was used as it was considered the optimal compression for this cell configuration. All carbon felt electrodes used in this set of experiments were thermally treated in air at 390 °C for 6 h.

Within the flow rate range tested, 150 $cm^3 min^{-1}$ (linear flow velocity of 1.26 $cm s^{-1}$) provided the highest storage capacity (*ca.* 2331 mA h) as a result of a longer time of charge and discharge cycle. Lowering the flow rate, the capacity also decreases: at 37.5 $cm^3 min^{-1}$ (0.32 $cm s^{-1}$) the discharge capacity of the VRFC is 1844 mA h. However, setting the flow rate to 300 $cm^3 min^{-1}$ (2.52 $cm s^{-1}$), a change in the trend is observed as the discharge capacity decreases 14% to 2004 mA h. Low flow rates (37.5 $cm^3 min^{-1}$) may cause an inefficient electrolyte circulation leading to stagnant regions inside the electrodes. As flow rate increases, the concentration polarization reduces improving mass transport phenomena between electrolyte and electrode leading to an increase of discharge capacity: 2331 mA h at 150 $cm^3 min^{-1}$. However, at the highest flow rate (300 $cm^3 min^{-1}$), the residence time is lower and the contact between the electrolyte and electrode is reduced, leading to a higher concentration polarization and, therefore to a reduction in the capacity [39,40]. In fact, Fig. 9b shows that CE decreased to 86% at 300 $cm^3 min^{-1}$. For 37 $cm^3 min^{-1}$, 75 $cm^3 min^{-1}$ and 150 $cm^3 min^{-1}$ the CE was 99%, 98% and 97% respectively, revealing a negative trend of this parameter. VE increased from 77% at 37.5 $cm^3 min^{-1}$ up to 87% at 150 and 300 $cm^3 min^{-1}$. M. R. Mohamed et al. [40], A. A. Shah et al. [41] also arrived to similar conclusions regarding flow rates and optimum cell efficiencies. The former concluded that flow rates higher than 180 $cm^3 min^{-1}$ (linear flow velocity of 1.11 $cm s^{-1}$) were no longer effective while below 180 $cm^3 min^{-1}$ the active species were not well distributed. In the latter study, the optimized flow rate was 120 $cm^3 min^{-1}$ as coulombic efficiency improves in the flow rate range of 60–120 $cm^3 min^{-1}$. However, the authors did not provide information regarding flow velocity.

Regarding DCL, Fig. 9b shows an almost linear increase with increasing flowrate in the range of 37.5–150 $cm^3 min^{-1}$: at 37.5 $cm^3 min^{-1}$, the VRFC lost 0.38% $cycle^{-1}$ while at 150 $cm^3 min^{-1}$ the loss was 1.8% $cycle^{-1}$. However, for 300 $cm^3 min^{-1}$ DCL increases dramatically to 3.5% $cycle^{-1}$. Several aspects may be considered when flow rate increases: *i*) full de-intercalation of SO_4^{2-}; *ii*) removal of SFG from electrode surface; *iii*) higher osmotic pressure in the membrane leading to higher net vanadium transfer the membrane and, consequently, electrolyte imbalance.

A suitable volumetric flow rate of electrolyte through each half-cell was found to be in the range 1.5–2.0 mL min^{-1}, corresponding to a mean linear electrolyte velocity of 1.0–10.1 $cm s^{-1}$ through the carbon felt electrode. A suitable volumetric flow rate of electrolyte through each half-cell was found to be in the range 1.5–2.0 mL min^{-1}, corresponding to a mean linear electrolyte velocity of 1.0–10.1 $cm s^{-1}$ through the carbon felt electrode.

The effect of electrolyte flow rate on the limiting current and on the R_{ohm} and R_{ct} is investigated in Fig. 10. R_{ct} does not change significantly with flow rate while R_{ohm} shows a slight decreasing trend – Fig. 10a – which should be related to a better distribution of the electrolyte in the electrode. Considering the high electrical conductivity of the electrolyte, a better distribution may result in lower contact resistances between the cell components.

Fig. 10b clearly shows the positive effect of the flow rate on the limiting current density: as the flow rate increases from 37.5 $cm^3 min^{-1}$ to 300 $cm^3 min^{-1}$ (corresponding to a mean linear velocity ranging between 0.29 and 2.34 $cm s^{-1}$) the limiting current density increases from 28 mA cm^{-2} to 210 mA cm^{-2}. Although it is not observed in the polarization curves (please see Appendix A), it is worth noting that at high current densities and high flow rates depletion of vanadium species from the electrolyte may occur [52]. In such case, limitations in the mass transport will become significant.

3.4. Influence of the current density

Fig. 11 shows the profile of the second of 20 charge/discharge cycles for different current densities (Fig. 11a) and efficiency and capacity fading histories (Fig. 11b). Charge/discharge tests were cycled between 1.60 V and 0.8 V under current densities of 40 mA cm^{-2} and 80 mA cm^{-2}. For 120 mA cm^{-2} and 160 mA cm^{-2} the tests were cycled between 1.70 V and 0.8 V and 1.75 V and 0.8 V, respectively. The difference in the voltage ranges were to compensate the overpotential losses at higher currents and to ensure the OCV between 1.2 and 1.5 V (SOC between 10 and 90%). To prevent electrode starvation due the complete depletion of the electrolyte at higher currents, 300 $cm^3 min^{-1}$ (linear flow velocity of 2.52 $cm s^{-1}$) was set. The charge/discharge time (*i.e.*, capacity) decreases dramatically as the current density increases: at 40 mA cm^{-2} the VRFC showed 2004 mA h of storage capacity whilst at 160 mA cm^{-2} a 36% capacity drop was registered corresponding to 1276 mA h (Fig. 11a). This reduction is a result of the increased overpotential caused by progressively higher charge and discharge rates, *i.e.*, current density, as it can be seen in Fig. 11a. Fig. 11b shows that CE increases from 87% at 40 mA cm^{-2} to 99% at 160 mA cm^{-2}.

Setting high flow rate at low current density (e.g. 40 mA cm^{-2}) not only can impair the reaction at the electrode, it can magnify the capacity loss as discussed in the previous sub-section. However, from Fig. 11b is possible to observe that increasing the current density for the same high flow rate (300 mL min^{-1}), the capacity loss decreases: at 40 mA cm^{-2} the capacity loss was ca. 3.5% $cycle^{-1}$ and above this value the fading is around 1.3% $cycle^{-1}$. High current densities (160 mA cm^{-2}) will induce lower charge/discharge times leading to a lower permeation of vanadium and consequently lower capacity fading. VE follows an opposite trend of CE: as the current density increases VE reduces from 87% to 55% at 40 mA cm^{-2} and 160 mA cm^{-2} respectively; this might be related to a significant increase of ohmic resistances and mass transport overpotentials as higher current densities are employed.

3.5. Influence of the membrane type

One of the most critical components of a VRFB is the membrane because it affects significantly the performance and cost of the VRFB system [53,54]. Therefore, three Nafion® membranes, specifically N115, N117 and, N212 (with as supplied thicknesses of 127 μm, 183 μm, and 51 μm, respectively) and the anionic membrane FAP 450 (55 μm as supplied thickness) were tested. The electrolyte was circulated at 150 $cm^3 min^{-1}$ (linear flow velocity of 1.26 $cm s^{-1}$).

Fig. 12 shows the influence of the membrane on the performance of the VRFC. Fig. 12a shows the second of 20 charge/discharge cycles at a current density of 40 mA cm^{-2}. The duration of the cycle increases following the order N212 < N115 < N117 < FAP 450 whereas capacity also follows the same trend, 1675 mA h < 2109 mA h < 2267 mA h < 2331 mA h. Under low current densities, increasing the Nafion® membrane thickness seems

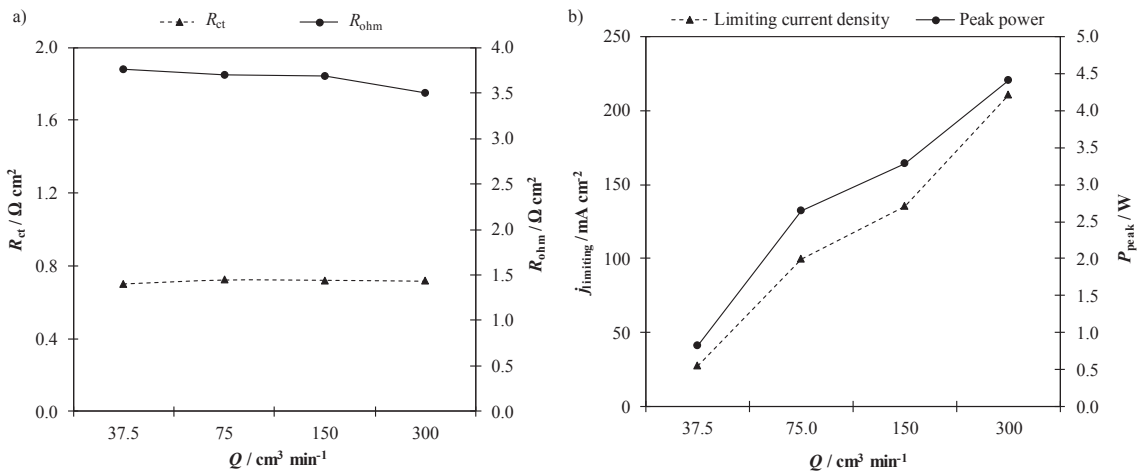

Fig. 10. Effect of $Q_{electrolyte}$ on: (a) R_{ct} and R_{ohm} at 50% SOC; (b) Limiting current density, $j_{limiting}$, and peak power, P_{peak}, between 1.6 and 0.8 V. Electrode compression; Electrodes treated in air atmosphere at 390 °C for 6 h; Membrane: FAP 450.

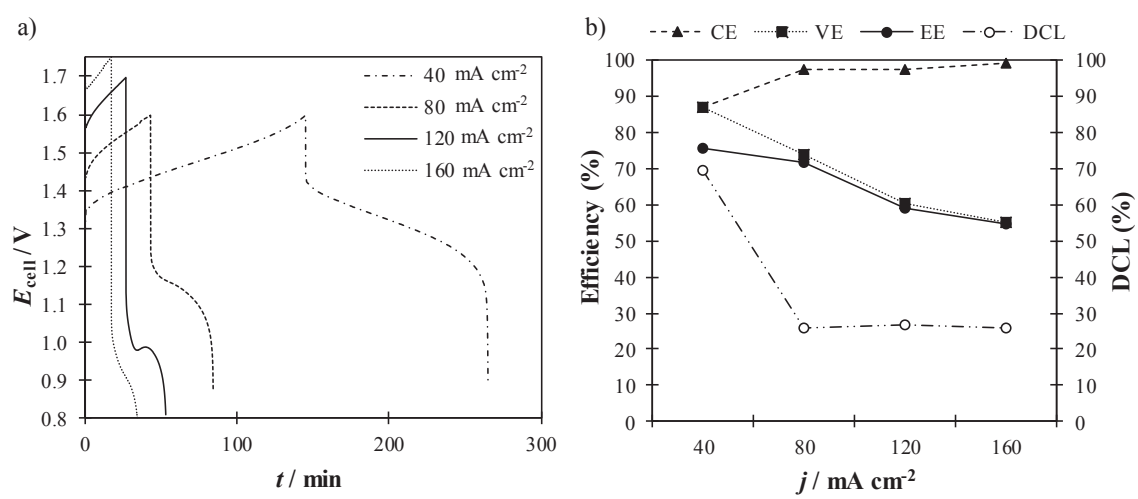

Fig. 11. Effect of current density, *j*, on: (a) the history of the second charge/discharge cycle; (b) average CE, VE, EE and DCL for 20 cycles. Electrode compression = 25%; $Q_{electrolyte}$ = 300 cm³ min⁻¹; Electrodes treated in air atmosphere at 390 °C for 6 h; Membrane: FAP 450. OCV_d = 1.2 V, OCV_c = 1.5 V.

advantageous as the crossover of vanadium species is reduced notwithstanding the increasing ohmic resistances. The crossover of vanadium through CEMs may not be avoided due to the interactions between negatively charge sulfonated groups from Nafion® and positively charged vanadium ions which may facilitate the adsorption and diffusion of vanadium ions [55]. Regarding the AEM tested, the crossover of vanadium species is expected to be drastically reduced due to the Donnan exclusion effect [56] that may justify the larger time of charge/discharge cycle for the AEM.

Fig. 12b illustrates the effect of the membrane type on the average efficiencies for the whole 20 cycles and on the capacity fading after 20 cycles of charge and discharge at 40 mA cm^{-2}. Reducing the thickness of the Nafion® membranes, higher rate of vanadium crossover is expected affecting negatively CE. VE was also the lowest for the N212 which may be related to higher vanadium crossover. The AEM, comparable to N212 in terms of thickness, yielded 97% of CE and 87% of VE due its repulsive properties towards vanadium ions, compensating the lower ionic conductivity of this membrane [54,55]. Regarding the storage capacity, Fig. 12 b. Shows that capacity fading also increases as Nafion® membranes thickness decreases (0.4% cycle^{-1}, 0.3% cycle^{-1}, 0.2% cycle^{-1} for N212, N115 and N117, respectively). However, capacity fading of FAP 450 was the highest: 1.8% cycle^{-1}. A possible explanation for such behaviour is the fact that although the diffusion of V^{2+} and V^{3+} co-ions is restricted, neutral $VOSO_4$ and negative $VO_2SO_4^-$ from the positive half-cell can easily permeate to the negative half-cell [57] leading to electrolyte imbalance. It is worth mentioning that CE and VE were rather stable, and no net water transfer was observed during the 20 cycles. Another possible explanation for the higher capacity loss of the AEM tested is the fact that most AEMs commercially available are used for alkaline electrolytes and are unstable in acidic media. Such degradation (chemical aging) may cause higher permeation of vanadium ions which more quickly imbalance the electrolyte and ultimately lead to higher losses in capacity [58–60].

Fig. 13 shows the variation of R_{ohm} and R_{ct} (Fig. 13a) and limiting current density and peak power (Fig. 13b) when the battery was equipped with the selected membranes. VRFC equipped with Nafion® showed increasing R_{ohm} and R_{ct} with increasing thicknesses, Fig. 13a. FAP 450 presented similar R_{ohm} and R_{ct} values as Nafion® 117 which may be related to its well-known lower anionic conductivity despite being much thinner than Nafion® 117 [59,61].

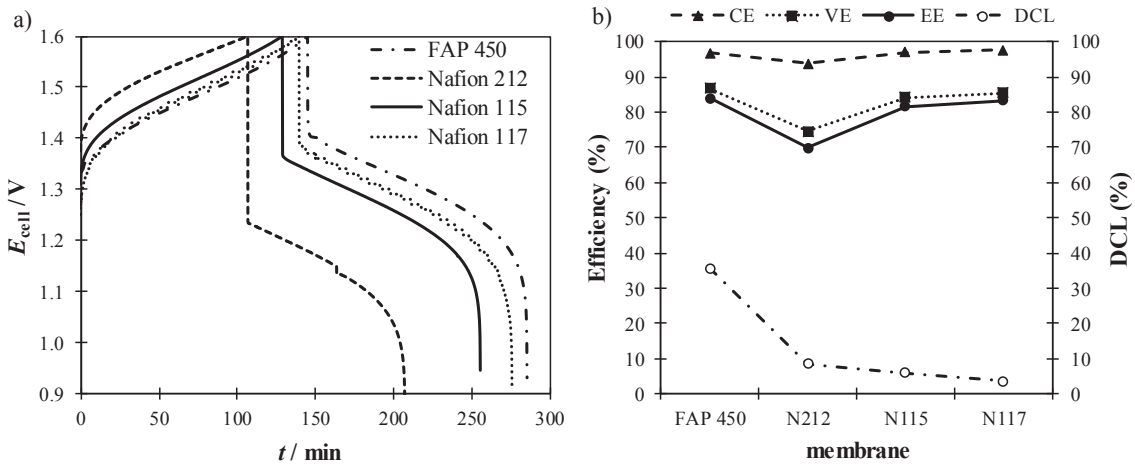

Fig. 12. Effect of membrane on: (a) the history of the second charge/discharge cycle; (b) average CE, VE, EE and DCL for 20 cycles. Electrodes treated in air atmosphere at 415 °C for 12 h; Electrode compression = 25%; $Q_{electrolyte}$ = 150 cm^3 min^{-1} j = 40 mA cm^{-2}; OCV$_d$ = 1.2 V, OCV$_c$ = 1.5 V.

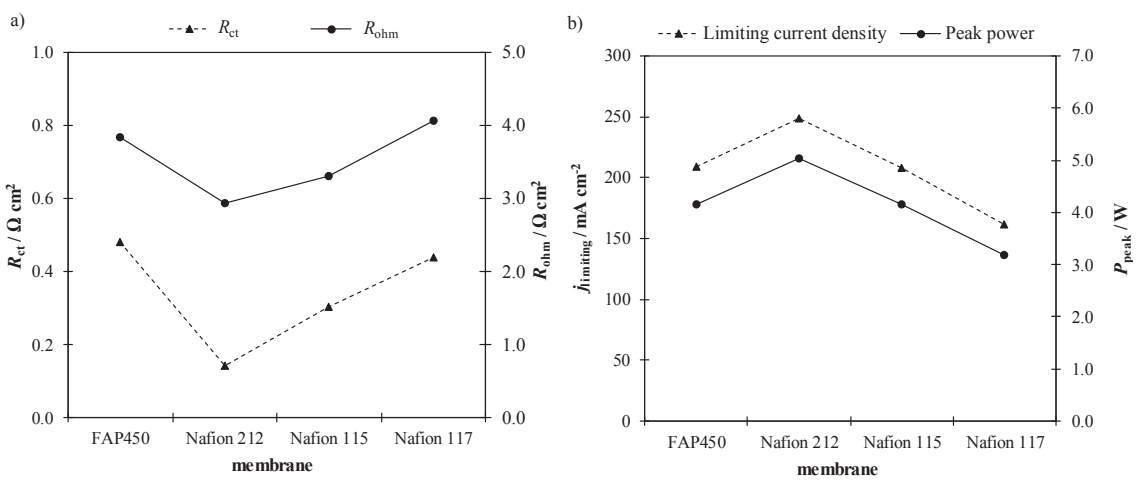

Fig. 13. Effect of the membrane on: (a) R_{ct} and R_{ohm} at 50% SOC; (b) Limiting current density, $j_{limiting}$, and peak power, P_{peak}, between 1.6 and 0.8 V. Electrodes compression = 25%; Electrodes treated in air at 415 °C for 12 h; $Q_{electrolyte}$ = 150 cm^3 min^{-1}.

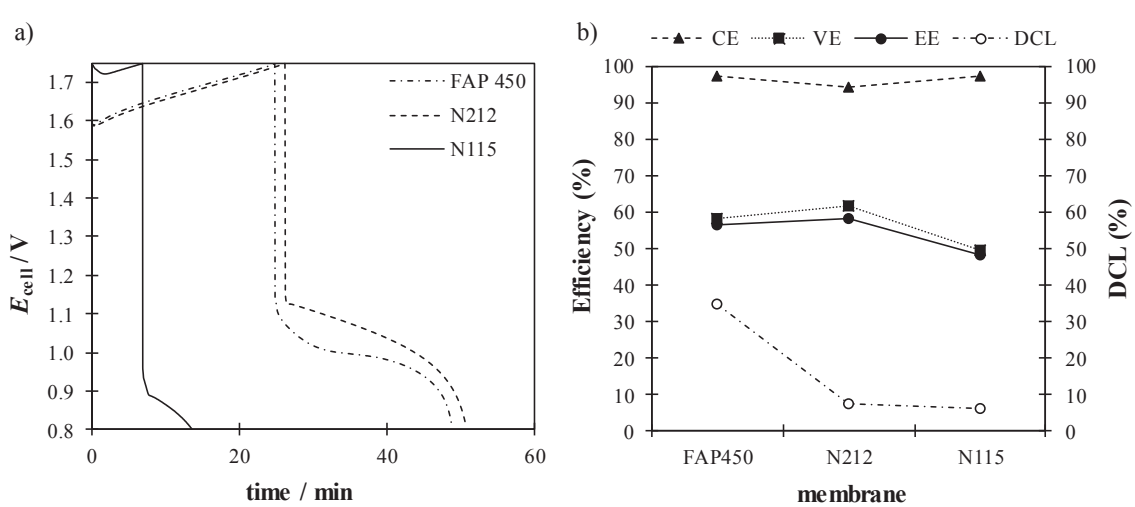

Fig. 14. Effect of membrane on: (a) the history of the second charge/discharge cycle; (b) CE, VE, EE and DCL after 20 cycles. Electrodes treated in air atmosphere at 415 °C for 12 h; Electrode compression = 25%; $Q_{electrolyte}$ = 150 cm^3 min^{-1} j = 160 mA cm^{-2}; OCV$_d$ = 1.2 V, OCV$_c$ = 1.5 V.

Regarding the limiting current density, as the Nafion® membrane thickness increases, the limiting current density decreases. This behaviour relates to the dominating ohmic and charge transfer losses as the membrane thickness increases. FAP 450 membrane showed similar limiting current density in comparison to Nafion® 115 – *ca.* 210 mA cm^{-2}.

Fig. 14 shows the effect of the same series of membranes on the VRFC performance but at 160 mA cm^{-2} of current density. As shown in Fig. 14a, N212 and FAP 450 based cell have similar storage capacities (*ca.* 1600 mA h). However, N115 shows lower storage capacity (414 mA h). In terms of energy efficiencies, N212 shows the highest VE (62%) while FAP 450 the highest CE (97%). The high value of VE found for N212 may be related to fact that at higher currents the vanadium crossover is no longer the dominate loss affecting the performance. The already mentioned Donnan exclusion effect [56,58] may justify the value found for CE when using FAP 450 membrane. The lower storage capacity and energy efficiencies found for N115 are related to the higher overpotential losses present at higher current densities. For this reason, N117, which is even thicker than N115, was not able to produce charge discharge cycles [10,31]. Similar values of storage capacity fading were found at 160 mA cm^{-2} compared to those obtained at 40 mA cm^{-2}, suggesting that high current densities have little effect on the capacity loss.

4. Conclusions

Within this work, the performance of a 25 cm^2 active area VRFC was successfully optimized. The thermal treatment condition of the carbon-based electrodes was systematically studied as well as the operating conditions such as electrode compression, flow rate and current density. In addition, the anionic membrane, FAP 450 was compared with Nafion® series.

The electrochemical activity of the electrodes may have two important contributions: 1) amount of oxygen functional groups present at the electrode surface and 2) surface morphological changes induced by the treatment. The optimum energy efficiency was obtained after thermal treating the carbon felt during 12 h at 390 °C and 415 °C in air; the energy efficiency increased from 51% (untreated electrode) to 85% at 40 mA cm^{-2}.

Although increasing the compression may reduce specific resistances due to a better electric contact between all components, the combined effect of ohmic resistance, electrolyte transport and distribution and charge/discharge time duration showed that 25% of compression yielded the highest value of EE - 84% at 40 mA cm^{-2}.

For the flow rate range tested in this study, 37.5 cm^3 min^{-1} to 300 cm^3 min^{-1}, the results showed that overpotential losses decreased as the flow rate increased whereas 150 cm^3 min^{-1} yielded the highest capacity (2331 mA h) and the highest EE (84%) under 40 mA cm^{-2}. However, high flow rates led to high capacity loss most likely due to the full de-intercalation of SO_4^{2-} ions from the electrode structure, removal of SFG from the fibres surface and to a higher osmotic pressure.

Increasing the current density from 40 mA cm^{-2} to 160 mA cm^{-2}, the capacity decreases dramatically, from 2004 mA h to 1276 mA h, due to the increased overpotential losses. Lower energy efficiencies are also obtained for higher current densities: under 40 mA cm^{-2} the electrochemical cell yielded 73% of energy efficiency while under 160 mA cm^{-2} only 54% was attained.

Nafion® membranes have been considered the benchmark for VRFB systems; however, FAP 450 yielded the highest EE and discharge capacity under 40 mA cm^{-2} – 84% and 2331 mA h, respectively. Under 160 mA cm^{-2}, FAP 450 yielded similar values of energy efficiency in comparison to N212: *ca.* 57%. A drawback of using membrane FAP 450 is the higher discharge capacity fading when compared to Nafion® membranes, however is much cheaper than Nafion® membranes.

Acknowledgements

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Appendix A. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.electacta.2018.02.054.

References

[1] Y.-K. Wu, S.-H. Tsai, M.-Y. Zou, Accommodating high PV penetration on the distribution system of Kinmen Island, Energy Power Eng. 5 (2013) 209–214.
[2] T. Shigematsu, Redox flow battery for energy storage, SEI Tech. Rev. 73 (2011) 4–13.
[3] C. Ponce de León, A. Frías-Ferrer, J. González-García, D.A. Szánto, F.C. Walsh, Redox flow cells for energy conversion, J. Power Sources 160 (2006) 716–732.
[4] A.Z. Weber, M.M. Mench, J.P. Meyers, P.N. Ross, J.T. Gostick, Q. Liu, Redox flow batteries: a review, J. Appl. Electrochem. 41 (2011) 1137–1164.
[5] A. Shibata, K. Sato, Development of vanadium redox flow battery for electricity storage, Power Eng. J. 13 (1999) 130–135.
[6] R. Ye, D. Henkensmeier, S.J. Yoon, Z. Huang, D.K. Kim, Z. Chang, S. Kim, R. Chen, Redox flow batteries for energy storage: a technology review, ASME J. Electrochem. Energy Convers. Storage 15 (2017), 010801.
[7] Á. Cunha, J. Martins, N. Rodrigues, F.P. Brito, Vanadium redox flow batteries: a technology review, Int. J. Energy Res. 39 (2015) 889–918.
[8] P. Leung, X. Li, C. Ponce de Leon, L. Berlouis, C.T.J. Low, F.C. Walsh, Progress in redox flow batteries, remaining challenges and their applications in energy storage, RSC Adv. 2 (2012) 10125–10156.
[9] P. Zhao, H. Zhang, H. Zhou, J. Chen, S. Gao, B. Yi, Characteristics and performance of 10kW class all-vanadium redox-flow battery stack, J. Power Sources 162 (2006) 1416–1420.
[10] D. Reed, E. Thomsen, W. Wang, Z. Nie, B. Li, X. Wei, B. Koeppel, V. Sprenkle, Performance of Nafion® N115, Nafion® NR-212, and Nafion® NR-211 in a 1 kW class all vanadium mixed acid redox flow battery, J. Power Sources 285 (2015) 425–430.
[11] X. Li, H. Zhang, Z. Mai, H. Zhang, I. Vankelecom, Ion exchange membranes for vanadium redox flow battery (VRB) applications, Energy Environ. Sci. 4 (2011) 1147–1160.
[12] B. Schwenzer, J. Zhang, S. Kim, L. Li, J. Liu, Z. Yang, Membrane development for vanadium redox flow batteries, ChemSusChem 4 (2011) 1388–1406.
[13] J. Xi, Z. Wu, X. Qiu, L. Chen, Nafion/SiO_2 hybrid membrane for vanadium redox flow battery, J. Power Sources 166 (2007) 531–536.
[14] J. Zeng, C. Jiang, Y. Wang, J. Chen, S. Zhu, B. Zhao, R. Wang, Studies on polypyrrole modified nafion membrane for vanadium redox flow battery, Electrochem. Commun. 10 (2008) 372–375.
[15] B. Schwenzer, S. Kim, M. Vijayakumar, Z. Yang, J. Liu, Correlation of structural differences between Nafion/polyaniline and Nafion/polypyrrole composite membranes and observed transport properties, J. Membr. Sci. 372 (2011) 11–19.
[16] J. Kim, J.-D. Jeon, S.-Y. Kwak, Nafion-based composite membrane with a permselective layered silicate layer for vanadium redox flow battery, Electrochem. Commun. 38 (2014) 68–70.
[17] Q. Luo, H. Zhang, J. Chen, P. Qian, Y. Zhai, Modification of Nafion membrane using interfacial polymerization for vanadium redox flow battery applications, J. Membr. Sci. 311 (2008) 98–103.
[18] J. Ma, S. Wang, J. Peng, J. Yuan, C. Yu, J. Li, X. Ju, M. Zhai, Covalently incorporating a cationic charged layer onto Nafion membrane by radiation-induced graft copolymerization to reduce vanadium ion crossover, Eur. Polym. J. 49 (2013) 1832–1840.
[19] H. Zhang, H. Zhang, X. Li, Z. Mai, J. Zhang, Nanofiltration (NF) membranes: the next generation separators for all vanadium redox flow batteries (VRBs)? Energy Environ. Sci. 4 (2011) 1676–1679.

[20] H. Zhang, H. Zhang, X. Li, Z. Mai, W. Wei, Silica modified nanofiltration membranes with improved selectivity for redox flow battery application, Energy Environ. Sci. 5 (2012) 6299–6303.
[21] S. Yun, J. Parrondo, V. Ramani, Derivatized cardo-polyetherketone anion exchange membranes for all-vanadium redox flow batteries, J. Mater. Chem. 2 (2014) 6605–6615.
[22] S. Zhang, B. Zhang, D. Xing, X. Jian, Poly(phthalazinone ether ketone ketone) anion exchange membranes with pyridinium as ion exchange groups for vanadium redox flow battery applications, J. Mater. Chem. 1 (2013) 12246–12254.
[23] H. Zhang, H. Zhang, F. Zhang, X. Li, Y. Li, I. Vankelecom, Advanced charged membranes with highly symmetric spongy structures for vanadium flow battery application, Energy Environ. Sci. 6 (2013) 776–781.
[24] L. Yue, W. Li, F. Sun, L. Zhao, L. Xing, Highly hydroxylated carbon fibres as electrode materials of all-vanadium redox flow battery, Carbon 48 (2010) 3079–3090.
[25] A. Di Blasi, N. Briguglio, O. Di Blasi, V. Antonucci, Charge–discharge performance of carbon fiber-based electrodes in single cell and short stack for vanadium redox flow battery, Appl. Energy 125 (2014) 114–122.
[26] A.M. Pezeshki, J.T. Clement, G.M. Veith, T.A. Zawodzinski, M.M. Mench, High performance electrodes in vanadium redox flow batteries through oxygen-enriched thermal activation, J. Power Sources 294 (2015) 333–338.
[27] O. Di Blasi, N. Briguglio, C. Busacca, M. Ferraro, V. Antonucci, A. Di Blasi, Electrochemical investigation of thermically treated graphene oxides as electrode materials for vanadium redox flow battery, Appl. Energy 147 (2015) 74–81.
[28] B. Sun, M. Skyllas-Kazacos, Modification of graphite electrode materials for vanadium redox flow battery application—I. Thermal treatment, Electrochim. Acta 37 (1992) 1253–1260.
[29] K.J. Kim, Y.-J. Kim, J.-H. Kim, M.-S. Park, The effects of surface modification on carbon felt electrodes for use in vanadium redox flow batteries, Mater. Chem. Phys. 131 (2011) 547–553.
[30] B. Sun, M. Skyllas-Kazakos, Chemical modification and electrochemical behaviour of graphite fibre in acidic vanadium solution, Electrochim. Acta 36 (1991) 513–517.
[31] Z. González, A. Sánchez, C. Blanco, M. Granda, R. Menéndez, R. Santamaría, Enhanced performance of a Bi-modified graphite felt as the positive electrode of a vanadium redox flow battery, Electrochem. Commun. 13 (2011) 1379–1382.
[32] C. Yao, H. Zhang, T. Liu, X. Li, Z. Liu, Carbon paper coated with supported tungsten trioxide as novel electrode for all-vanadium flow battery, J. Power Sources 218 (2012) 455–461.
[33] K.J. Kim, M.-S. Park, J.-H. Kim, U. Hwang, N.J. Lee, G. Jeong, Y.-J. Kim, Novel catalytic effects of Mn_3O_4 for all vanadium redox flow batteries, Chem. Commun. 48 (2012) 5455–5457.
[34] B. Li, M. Gu, Z. Nie, X. Wei, C. Wang, V. Sprenkle, W. Wang, Nanorod niobium oxide as powerful catalysts for an all vanadium redox flow battery, Nano Lett. 14 (2014) 158–165.
[35] T.-C. Chang, J.-P. Zhang, Y.-K. Fuh, Electrical, mechanical and morphological properties of compressed carbon felt electrodes in vanadium redox flow battery, J. Power Sources 245 (2014) 66–75.
[36] S.-K. Park, J. Shim, J.H. Yang, C.-S. Jin, B.S. Lee, Y.-S. Lee, K.-H. Shin, J.-D. Jeon, The influence of compressed carbon felt electrodes on the performance of a vanadium redox flow battery, Electrochim. Acta 116 (2014) 447–452.
[37] Q.H. Liu, G.M. Grim, A.B. Papandrew, A. Turhan, T.A. Zawodzinski, M.M. Mench, High performance vanadium redox flow batteries with optimized electrode Configuration and membrane selection, J. Electrochem. Soc. 159 (2012) A1246–A1252.
[38] M.J. Watt-Smith, P. Ridley, R.G.A. Wills, A.A. Shah, F.C. Walsh, The importance of key operational variables and electrolyte monitoring to the performance of an all vanadium redox flow battery, J. Chem. Technol. Biotechnol. 88 (2013) 126–138.
[39] X. Ma, H. Zhang, C. Sun, Y. Zou, T. Zhang, An optimal strategy of electrolyte flow rate for vanadium redox flow battery, J. Power Sources 203 (2012) 153–158.
[40] M.R. Mohamed, P.K. Leung, M.H. Sulaiman, Performance characterization of a vanadium redox flow battery at different operating parameters under a standardized test-bed system, Appl. Energy 137 (2015) 402–412.
[41] A.A. Shah, M.J. Watt-Smith, F.C. Walsh, A dynamic performance model for redox-flow batteries involving soluble species, Electrochim. Acta 53 (2008) 8087–8100.
[42] H.A.-Z. Al-Fetlawi, The electrolyte monitoring of a vanadium redox flow battery, ECS Transactions 58 (2014) 33–48.
[43] N.R. Laine, F.J. Vastola, P.L. Walker, The importance of active surface area in the carbon-oxygen reaction, J. Phys. Chem. 67 (1963) 2030–2034.
[44] P.J. Hart, F.J. Vastola, P.L. Walker, Oxygen chemisorption on well cleaned carbon surfaces, Carbon 5 (1967) 363–371.
[45] V.A. Sethuraman, L.J. Hardwick, V. Srinivasan, R. Kostecki, Surface structural disordering in graphite upon lithium intercalation/deintercalation, J. Power Sources 195 (2010) 3655–3660.
[46] A. Moissette, H. Fuzellier, A. Burneau, J. Dubessy, M. Lelaurain, Sulfate graphite intercalation compounds: new electrochemical data and spontaneous intercalation, Carbon 33 (1995) 123–128.
[47] I. Derr, M. Bruns, J. Langner, A. Fetyan, J. Melke, C. Roth, Degradation of all-vanadium redox flow batteries (VRFB) investigated by electrochemical impedance and X-ray photoelectron spectroscopy: Part 2 electrochemical degradation, J. Power Sources 325 (2016) 351–359.
[48] S.M. Park, H. Kim, Porous electrodes with lower impedance for vanadium redox flow batteries, Korean Chem. Eng. Res. 53 (2015) 638–645.
[49] K. Yoshizawa, K. Ikezoe, Y. Tasaki, D. Kramer, E.H. Lehmann, G.G. Scherer, Analysis of gas diffusion layer and flow-field design in a PEMFC using neutron radiography, J. Electrochem. Soc. 155 (2008) B223–B227.
[50] A. Tang, J. Bao, M. Skyllas-Kazacos, Dynamic modelling of the effects of ion diffusion and side reactions on the capacity loss for vanadium redox flow battery, J. Power Sources 196 (2011) 10737–10747.
[51] A. Tang, J. Bao, M. Skyllas-Kazacos, Studies on pressure losses and flow rate optimization in vanadium redox flow battery, J. Power Sources 248 (2014) 154–162.
[52] D. Aaron, Z. Tang, A.B. Papandrew, T.A. Zawodzinski, Polarization curve analysis of all-vanadium redox flow batteries, J. Appl. Electrochem. 41 (2011) 1175–1182.
[53] B. Jiang, L. Wu, L. Yu, X. Qiu, J. Xi, A comparative study of Nafion series membranes for vanadium redox flow batteries, J. Membr. Sci. 510 (2016) 18–26.
[54] D. Chen, M.A. Hickner, E. Agar, E.C. Kumbur, Selective anion exchange membranes for high coulombic efficiency vanadium redox flow batteries, Electrochem. Commun. 26 (2013) 37–40.
[55] Z. Mai, H. Zhang, H. Zhang, W. Xu, W. Wei, H. Na, X. Li, Anion-conductive membranes with ultralow vanadium permeability and excellent performance in vanadium flow batteries, ChemSusChem 6 (2013) 328–335.
[56] S. Sarkar, A.K. SenGupta, P. Prakash, The donnan membrane principle: opportunities for sustainable engineered processes and materials, Environ. Sci. Technol. 44 (2010) 1161–1166.
[57] T. Mohammadi, S.C. Chieng, M. Skyllas Kazacos, Water transport study across commercial ion exchange membranes in the vanadium redox flow battery, J. Membr. Sci. 133 (1997) 151–159.
[58] P.K. Leung, Q. Xu, T.S. Zhao, L. Zeng, C. Zhang, Preparation of silica nanocomposite anion-exchange membranes with low vanadium-ion crossover for vanadium redox flow batteries, Electrochim. Acta 105 (2013) 584–592.
[59] H. Prifti, A. Parasuraman, S. Winardi, T.M. Lim, M. Skyllas-Kazacos, Membranes for redox flow battery applications, Membranes 2 (2012) 275.
[60] S. Slade, S.A. Campbell, T.R. Ralph, F.C. Walsh, Ionic conductivity of an extruded nafion 1100 EW series of membranes, J. Electrochem. Soc. 149 (2002) A1556–A1564.
[61] J. Fang, H. Xu, X. Wei, M. Guo, X. Lu, C. Lan, Y. Zhang, Y. Liu, T. Peng, Preparation and characterization of quaternized poly (2,2,2-trifluoroethyl methacrylate-co-N-vinylimidazole) membrane for vanadium redox flow battery, Polym. Adv. Technol. 24 (2013) 168–173.